

Question:

Carlos is planting a lemon tree. The tree will cost \$90 to plant. Each year it will grow 7 lemons, which he can sell for \$1.5 each. It costs \$3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

$\mathbb{D}\mu\mathbb{D}\mathbb{I}\mathbb{D}\mu$ $\mathbb{D}^{\frac{1}{2}}\tilde{\mathbf{N}}\mathbb{H}\mathbb{D}\mathbb{I}\mathbb{D}^{\frac{1}{2}}\mathbb{D}^{\frac{3}{4}}$ $\mathbb{D}^{\frac{3}{4}}\mathbb{D}_{\mathbb{Z}}$ $\tilde{\mathbf{N}}\mathbb{G}\mathbb{D}\mu\mathbb{D}'\mathbb{D}\mu\mathbb{D}\gg\mathbb{D}_{\mathbb{Z}}\tilde{\mathbf{N}}\hat{\mathbf{N}}\mathbb{I}$, ****** $\mathbb{D}^{\circ}\mathbb{D}^{\frac{3}{4}}\mathbb{D}^3\mathbb{D}'\mathbb{D}^{\circ}$ Carlos $\mathbb{D}^{\frac{1}{2}}\mathbb{D}^{\circ}\tilde{\mathbf{N}}\mathbb{I}\mathbb{D}_{\mathbb{D}^{\frac{1}{2}}\mathbb{D}^{\circ}}$ $\mathbb{D}\mu\tilde{\mathbf{N}}\hat{\mathbf{H}}$
 $\mathbb{D}\cdot\mathbb{D}^{\circ}$ $\tilde{\mathbf{N}}\mathbb{G}\mathbb{D}^{\circ}\mathbb{D}^{\pm}\mathbb{D}^{\circ}\tilde{\mathbf{N}}\hat{\mathbf{H}}\tilde{\mathbf{N}}\mathbb{I}\mathbb{D}^2\mathbb{D}^{\circ}\tilde{\mathbf{N}}\hat{\mathbf{H}}\mathbb{I}$ $\mathbb{D}'\mathbb{D}\mu\mathbb{D}^{\frac{1}{2}}\mathbb{N}\mathbb{I}\mathbb{D}^3\mathbb{D}_{\mathbb{Z}}$ ****** , $\tilde{\mathbf{N}}\hat{\mathbf{H}}\mathbb{D}^{\frac{3}{4}}$ $\mathbb{D}\mu\tilde{\mathbf{N}}\acute{\mathbf{G}}\tilde{\mathbf{N}}\hat{\mathbf{H}}\mathbb{I}$ $\mathbb{D}^{\circ}\mathbb{D}^{\frac{3}{4}}\mathbb{D}^3\mathbb{D}'\mathbb{D}^{\circ}$ $\mathbb{D}\mu\mathbb{D}^3\mathbb{D}^{\frac{3}{4}}$ ****** $\mathbb{D}'\mathbb{D}^{\frac{3}{4}}$
 $\tilde{\mathbf{N}}\mathbb{H}\mathbb{D}^{\frac{3}{4}}\mathbb{D}'$ $\tilde{\mathbf{N}}\mathbb{I}$ $\mathbb{D}_{\mathbb{Z}}$ $\tilde{\mathbf{N}}\mathbb{G}\mathbb{D}\mu\mathbb{D}^2$ $\tilde{\mathbf{N}}\mathbb{I}\tilde{\mathbf{N}}\mathbb{I}\mathbb{D}^{\circ}\tilde{\mathbf{N}}\mathbb{I}\tilde{\mathbf{N}}\hat{\mathbf{H}}$ $\tilde{\mathbf{N}}\mathbb{G}\mathbb{D}^{\circ}\tilde{\mathbf{N}}\acute{\mathbf{G}}\tilde{\mathbf{N}}\mathbb{H}\mathbb{D}^{\frac{3}{4}}\mathbb{D}'$ $\tilde{\mathbf{N}}\mathbb{I}$ ****** .

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$\mathbb{D} \mathbb{K} \mathbb{D}^0 \mathbb{D}^{1/2} \mathbb{D}^{3/4}$:

- $\partial_i \tilde{N} \hat{D}^{3/4} \partial_j D^{1/4} \tilde{D}^{3/4} \tilde{N} \hat{g} \tilde{N} \hat{H} \tilde{D} \partial \partial^{3/4} \tilde{N} \hat{g} D' \partial^0 D_j : \$ 90 (\tilde{N}_i \tilde{N} \hat{D}^{3/4} \partial^{3/4} D' \partial_j D^{1/2} \tilde{N} \hat{g} D' D \cdot , ** D^2$
 $\partial^{1/2} D^0 \tilde{N} \hat{D}^0 D \gg \partial_\mu (**)$

- $\partial_i \partial^0 D \partial \partial \tilde{N} \hat{D}^1 \partial^3 D^{3/4} D' :$

- $\partial_i \partial^{3/4} \tilde{N} \hat{g} \tilde{N} \hat{H} 7 D \gg \partial_j D^{1/4} \partial^{3/4} D^{1/2} \partial^{3/4} D^2$

- $\partial_i \tilde{N} \hat{D}^{3/4} \partial_j D^{1/4} \partial^{3/4} \tilde{N} \hat{g} \tilde{N} \hat{H} \tilde{N}_i 1 D \gg \partial_j D^{1/4} \partial^{3/4} D^{1/2} D^0 : \$ 1.5$

- $\partial \hat{L} D^0 \tilde{N} \hat{H} \tilde{N} \hat{g} D^0 \tilde{N} \hat{H} \tilde{N}_i \partial^{1/2} D^0 \tilde{N} \hat{h} \tilde{N} \hat{h} \partial^{3/4} D' : \$ 3 D^2 \partial^3 D^{3/4} D'$

Đ°Đ³¹: ĐĐ³¼ÑĥĐ³¼Đ'ÑĭĐ, ÑĠĐ°ÑġÑĥĐ³¼Đ'ÑĭĐ²Đ³Đ³¼Đ'

- $\text{DK}\mathbb{D}^{\frac{3}{4}}\tilde{\text{N}}\mathbb{h}\mathbb{D}^{\frac{3}{4}}\mathbb{D}'\tilde{\text{N}}: 7\mathbb{D}\gg\mathbb{D}_\text{J}\mathbb{D}^{\frac{1}{4}}\mathbb{D}^{\frac{3}{4}}\mathbb{D}^{\frac{1}{2}}\mathbb{D}^{\frac{3}{4}}\mathbb{D}^2\tilde{\text{A}}\text{L}\$1.5 = \$10.5$
 - $\text{D}\mathbb{h}\mathbb{D}^\circ\tilde{\text{N}}\mathbb{g}\tilde{\text{N}}\mathbb{h}\mathbb{D}^{\frac{3}{4}}\mathbb{D}'\tilde{\text{N}}: \$3(\mathbb{D}^{\frac{1}{2}}\mathbb{D}^\circ\tilde{\text{N}}\mathbb{h}\tilde{\text{N}}\mathbb{h}\mathbb{D}^{\frac{3}{4}}\mathbb{D}')$
 - $\text{D}\text{L}\tilde{\text{N}}\mathbb{G}\mathbb{D}_\text{J}\mathbb{D}\pm\tilde{\text{N}}\mathbb{D}\gg\tilde{\text{N}}\mathbb{D}^2\mathbb{D}^3\mathbb{D}^{\frac{3}{4}}\mathbb{D}': \$10.5 - \$3 = \7.5

[illegible][illegible][illegible]
$$\mathfrak{D}\mathfrak{c}\mathfrak{D}^{3/4}\mathfrak{D}_{\mu}\tilde{\mathfrak{N}}\mathfrak{g}\tilde{\mathfrak{N}}\hat{\mathfrak{H}}\tilde{\mathfrak{N}}_{\mathfrak{l}},\mathfrak{D}^{\circ}\mathfrak{D}^{3/4}\mathfrak{D}^3\mathfrak{D}'\mathfrak{D}^{\circ}:$$

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> $7.5 \times n > 90$
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$$\mathbb{D}^3 \mathbb{D}' \mathbb{D}_\mu \text{ n } \hat{\mathbb{G}} \mathbb{K} \mathbb{D}^\circ \mathbb{D}^{3/4} \mathbb{D} \gg \mathbb{D}_\text{,} \tilde{\mathbb{N}} \tilde{\mathbb{D}}_\mu \tilde{\mathbb{N}} \hat{\mathbb{G}} \tilde{\mathbb{N}} \hat{\mathbb{H}} \mathbb{D}^2 \mathbb{D}^{3/4} \mathbb{D} \gg \mathbb{D}_\mu \tilde{\mathbb{N}} \hat{\mathbb{H}} .$$

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$\mathcal{D} \vdash \mathcal{D}_{\mu\tilde{N}\bar{I}} \mathcal{D}_{\cdot} \mathcal{D}^{1/4} :$

$$\begin{aligned} \$7.5n &> 90\$ \\ \$n &> 90 / 7.5\$ \\ \$n &> 12\$ \end{aligned}$$

$\mathbb{D}^{\frac{1}{2}}\mathbb{D}^{\circ}\tilde{\mathbb{N}}\tilde{\mathbb{D}}_{\cdot}\tilde{\mathbb{N}}\hat{\mathbb{H}}, \quad **\tilde{\mathbb{N}}\tilde{\mathbb{D}}\mu\tilde{\mathbb{N}}\mathbb{G}\mathbb{D}\mu\mathbb{D}\cdot 12 \quad \mathbb{D}\gg\mathbb{D}\mu\tilde{\mathbb{N}}\hat{\mathbb{H}} \quad **\mathbb{D}^3\mathbb{D}^{\frac{1}{2}} \quad **\tilde{\mathbb{N}}\hat{\mathbb{H}}\mathbb{D}^3\mathbb{D}\gg\tilde{\mathbb{N}}\mathbb{I}\mathbb{D}^{\circ}\mathbb{D}^3\mathbb{D} \quad \tilde{\mathbb{N}}\tilde{\mathbb{N}}\hat{\mathbb{H}}\mathbb{D}^3\mathbb{D} \quad \mathbb{D}^3\mathbb{D}^{\circ}\tilde{\mathbb{N}}\hat{\mathbb{h}}\mathbb{D}_{\cdot}$
 $\mathbb{D}_{\cdot}\tilde{\mathbb{N}}\hat{\mathbb{H}} \quad **\mathbb{D}\cdot\mathbb{D}^{\circ}\tilde{\mathbb{N}}\hat{\mathbb{H}}\tilde{\mathbb{N}}\mathbb{G}\mathbb{D}^{\circ}\tilde{\mathbb{N}}\hat{\mathbb{H}}\tilde{\mathbb{N}}\mathbb{I} \quad (\tilde{\mathbb{N}}\hat{\mathbb{H}}\mathbb{D}^{\circ}\mathbb{D}^{\circ} \quad \mathbb{D}^{\circ}\mathbb{D}^{\circ}\mathbb{D}^{\circ} \quad \tilde{\mathbb{N}}\tilde{\mathbb{D}}\mu\tilde{\mathbb{N}}\mathbb{G}\mathbb{D}\mu\mathbb{D}\cdot 12 \quad \mathbb{D}\gg\mathbb{D}\mu\tilde{\mathbb{N}}\hat{\mathbb{H}} \quad \mathbb{D}_{\cdot}\mathbb{D}^3\mathbb{D}\gg\tilde{\mathbb{N}}\hat{\mathbb{h}}\tilde{\mathbb{N}}\mathbb{I}\tilde{\mathbb{D}}_{\cdot}\tilde{\mathbb{N}}\hat{\mathbb{H}} \quad \mathbb{D}_{\cdot}\tilde{\mathbb{N}}\mathbb{G}\mathbb{D}_{\cdot}$
 $\mathbb{D}\pm\tilde{\mathbb{N}}\tilde{\mathbb{D}}\gg\tilde{\mathbb{N}}\mathbb{I} \quad 12 \quad \tilde{\mathbb{A}}\tilde{\mathbb{L}} \quad 7.5 = 90, \quad \tilde{\mathbb{N}}\hat{\mathbb{H}}\mathbb{D}^3\mathbb{D} \quad \mathbb{D}\mu\tilde{\mathbb{N}}\mathbb{G}\tilde{\mathbb{N}}\hat{\mathbb{H}}\tilde{\mathbb{N}}\mathbb{I} \quad **\tilde{\mathbb{N}}\hat{\mathbb{H}}\mathbb{D}^3\mathbb{D}\gg\tilde{\mathbb{N}}\mathbb{I}\mathbb{D}^{\circ}\mathbb{D}^3\mathbb{D} \quad \mathbb{D}^3\mathbb{D}^{\circ}\tilde{\mathbb{N}}\hat{\mathbb{h}}\mathbb{D}_{\cdot} \quad \mathbb{D}^{\circ}\mathbb{D}\mu\tilde{\mathbb{N}}\hat{\mathbb{H}} \quad **, \quad \mathbb{D}^{\frac{1}{2}}\mathbb{D}^3\mathbb{D} \quad **$
 $\mathbb{D}^{\frac{1}{2}}\mathbb{D}^{\circ}\tilde{\mathbb{N}}\tilde{\mathbb{D}}_{\cdot}\mathbb{D}^{\frac{1}{2}}\mathbb{D}^{\circ} \quad \mathbb{D}\mu\tilde{\mathbb{N}}\hat{\mathbb{H}} \quad \mathbb{D}\cdot\mathbb{D}^{\circ} \quad \tilde{\mathbb{N}}\mathbb{G}\mathbb{D}^{\circ}\mathbb{D}\pm\mathbb{D}^{\circ}\tilde{\mathbb{N}}\hat{\mathbb{H}} \quad \tilde{\mathbb{N}}\tilde{\mathbb{D}}^2\mathbb{D}^{\circ}\tilde{\mathbb{N}}\hat{\mathbb{H}}\tilde{\mathbb{N}}\mathbb{I} \quad ** \quad \tilde{\mathbb{N}}\mathbb{G} \quad 13 - \mathbb{D}^3\mathbb{D}^3\mathbb{D} \quad \mathbb{D}^3\mathbb{D}^3\mathbb{D}^{\frac{1}{2}} \quad \mathbb{D}^{\circ}).$

> ÐŁ Ð³¼Ñİ Ğğ Ð½ ÐµÐ½Ð½ Ðµ : Ð² 1 2 - Ð¼ Ð³Ð³¼Ð´ Ğĥ Ð³¼Ð½ ĞĥÐ¶Ðµ Ð² Ð³¼Ð·Ð²ĞÐ°ÑİÐ°ÐµÑĤ 1 2
 ĀĹ 7 . 5 = \$ 9 0 , ĞĥÐ³¼ ÐµÑğÑĤÑĹ ** Ð² ĞĥÐ³¼Ñİ Ð½ Ð³¼ÑğÑĤÐ , Ð³¼ Ð°ÑĥÐġ Ð· Ð» ** ÐġÐµÑĞÐ² Ð³¼
 Ð½Ð²Ð°ÑİÐ°Ð»Ñİ Ð½ÑİÐµ Ð· Ð°ÑĤ ĞĞÐ°ÑĤ Ğİ .

$\triangleright \mathbb{D}^{\mathbb{L}} \mathbb{D}^{\frac{3}{4}} \mathbb{D}^{\frac{3}{4}} \mathbb{D}^{\frac{1}{2}} ** \mathbb{D}^{\frac{1}{2}} \mathbb{D}^{\circ} \tilde{\mathbb{N}} \mathbb{D} \mathbb{D}^{\frac{1}{2}} \mathbb{D}^{\circ} \mathbb{D} \mu \tilde{\mathbb{N}} \hat{\mathbb{H}} \mathbb{D} \cdot \mathbb{D}^{\circ} \tilde{\mathbb{N}} \mathbb{G} \mathbb{D}^{\circ} \mathbb{D} \pm \mathbb{D}^{\circ} \tilde{\mathbb{N}} \hat{\mathbb{H}} \tilde{\mathbb{N}} \mathbb{D}^2 \mathbb{D}^{\circ} \tilde{\mathbb{N}} \hat{\mathbb{H}} \tilde{\mathbb{N}} ** ** \tilde{\mathbb{N}} \mathbb{G} \mathbb{D}^{\frac{1}{4}} \mathbb{D}^{\frac{3}{4}} \mathbb{D}^{\frac{1}{4}}$
 $\mathbb{D} \mu \mathbb{D}^{\frac{1}{2}} \tilde{\mathbb{N}} \hat{\mathbb{H}} \mathbb{D}^{\circ} , \mathbb{D}^{\circ} \mathbb{D}^{\frac{3}{4}} \mathbb{D}^3 \mathbb{D}^{\circ} \mathbb{D}^{\circ} \mathbb{D}^{\frac{3}{4}} \mathbb{D} \pm \tilde{\mathbb{N}} \mathbb{I} \mathbb{D} \mathbb{D} \mu \mathbb{D}^{\circ} \mathbb{D}^{\frac{3}{4}} \tilde{\mathbb{N}} \mathbb{h} \mathbb{D}^{\frac{3}{4}} \mathbb{D}^{\circ} \tilde{\mathbb{N}} \mathbb{I} \mathbb{D} \mathbb{D} \mathbb{G} \mathbb{D} \mu \mathbb{D}^2 \tilde{\mathbb{N}} \mathbb{I} \tilde{\mathbb{N}} \mathbb{I} \mathbb{D}^{\circ} \tilde{\mathbb{N}} \mathbb{I} \hat{\mathbb{N}} \hat{\mathbb{H}} \tilde{\mathbb{N}} \mathbb{G} \mathbb{D}^{\circ} \tilde{\mathbb{N}} \mathbb{G} \tilde{\mathbb{N}} \mathbb{h} \mathbb{D}^{\frac{3}{4}} \mathbb{D}^{\circ}$
 $\tilde{\mathbb{N}} \mathbb{I} ** \hat{\mathbb{a}} \mathbb{G} \mathbb{K} \tilde{\mathbb{N}} \hat{\mathbb{H}} \mathbb{D}^{\frac{3}{4}} \mathbb{D} \mu \tilde{\mathbb{N}} \mathbb{G} \tilde{\mathbb{N}} \hat{\mathbb{H}} \tilde{\mathbb{N}} ** \mathbb{D} \mathbb{D} \mathbb{D}^{\frac{3}{4}} \tilde{\mathbb{N}} \mathbb{G} \mathbb{D} \gg \mathbb{D} \mu ** 12 - \mathbb{D}^3 \mathbb{D}^{\frac{3}{4}} \mathbb{D}^3 \mathbb{D}^{\frac{3}{4}} \mathbb{D}^{\circ} \mathbb{D}^{\circ} , \mathbb{D}^{\circ} \mathbb{D}^{\frac{3}{4}} \mathbb{D}^3 \mathbb{D}^{\circ} \mathbb{D}^{\circ} \mathbb{D} \mu \mathbb{D}^3 \mathbb{D}^{\frac{3}{4}}$
 $\mathbb{D} \mathbb{D} \mathbb{G} \mathbb{D} \mathbb{D} \pm \tilde{\mathbb{N}} \mathbb{I} \mathbb{D} \gg \tilde{\mathbb{N}} \mathbb{I} \tilde{\mathbb{N}} \mathbb{G} \tilde{\mathbb{N}} \hat{\mathbb{H}} \mathbb{D}^{\circ} \mathbb{D}^{\frac{1}{2}} \mathbb{D}^{\frac{3}{4}} \mathbb{D}^2 \mathbb{D} \mathbb{D} \tilde{\mathbb{N}} \hat{\mathbb{H}} \tilde{\mathbb{N}} \mathbb{G} \tilde{\mathbb{N}} \mathbb{I} \mathbb{D} \mathbb{D}^{\frac{3}{4}} \mathbb{D} \gg \mathbb{D}^{\frac{3}{4}} \mathbb{D} \mathbb{I} \mathbb{D} \mathbb{D} \tilde{\mathbb{N}} \hat{\mathbb{H}} \mathbb{D} \mu \mathbb{D} \gg \tilde{\mathbb{N}} \mathbb{I} \mathbb{D}^{\frac{1}{2}} \mathbb{D}^{\frac{3}{4}} \mathbb{D}^1 \mathbb{D} \mathbb{D} ** \mathbb{D}^{\frac{3}{4}} \mathbb{D}^{\frac{1}{2}}$
 $\tilde{\mathbb{N}} \mathbb{h} \mathbb{D} \mathbb{I} \mathbb{D} \mu \mathbb{D}^{\frac{1}{2}} \mathbb{D} \mu \tilde{\mathbb{N}} \hat{\mathbb{H}} \mathbb{D} \mu \tilde{\mathbb{N}} \mathbb{G} \tilde{\mathbb{N}} \mathbb{I} \mathbb{D} \mu \tilde{\mathbb{N}} \hat{\mathbb{H}} \mathbb{D}^{\circ} \mathbb{D} \mu \mathbb{D}^{\frac{1}{2}} \tilde{\mathbb{N}} \mathbb{I} \mathbb{D}^3 \mathbb{D} ** .$

ĐŁ ÑĠĐ^{3/4}Đ²Đ_μÑĠĐ_ςĐ^{1/4}:

- $\partial \S \partial \mu \tilde{N} \zeta \partial \mu \partial \cdot 1 \partial^3 \partial^3 \partial^4 \partial^{\cdot} : \$ 7.5 \hat{\alpha} \zeta \zeta \tilde{N} \hat{\partial} \pm \tilde{N} \hat{N} \hat{H} \partial^3 \partial^4 \partial^{\circ}, \partial^2 \tilde{N} \acute{g} \partial \mu \partial^3 \partial^3 \partial^4 \partial \cdot \partial^{\circ} \hat{N} \hat{H} \tilde{N} \zeta \partial^{\circ} \hat{N} \hat{H} \tilde{N} : \$ 90$
- $\partial \S \partial \mu \tilde{N} \zeta \partial \mu \partial \cdot 12 \partial \gg \partial \mu \hat{N} \hat{H} : 12 \tilde{A} \tilde{L} \$ 7.5 = \$ 90 \hat{\alpha} \tilde{I} \tilde{J} ** \partial^2 \partial^3 \partial^4 \partial \cdot \partial^2 \tilde{N} \zeta \partial^{\circ} \tilde{N} \hat{I} \partial \cdot \partial \mu \hat{N} \hat{H} \tilde{N} \acute{g} \tilde{N} \hat{I} \partial \cdot \partial \mu \tilde{N} \zeta \partial^2 \partial^3 \partial^4$
 $\partial^{1/2} \partial^{\circ} \tilde{N} \hat{I} \partial^{\circ} \partial \gg \tilde{N} \hat{I} \partial^{1/2} \tilde{N} \hat{I} \partial \mu \partial \cdot \partial^{\circ} \hat{N} \hat{H} \tilde{N} \zeta \partial^{\circ} \hat{N} \hat{H} \tilde{N} ** , \partial \cdot \tilde{N} \zeta \partial \cdot \partial \pm \tilde{N} \hat{I} \partial \gg \tilde{N} \hat{I} = 0$
- $\partial \cdot \partial \gg \partial \mu \partial^{\cdot} \partial^3 \partial^4 \partial^2 \partial^{\circ} \hat{N} \hat{H} \partial \mu \partial \gg \tilde{N} \hat{I} \partial^{1/2} \partial^3 \partial^4 , ** \tilde{N} \acute{g} 13 - \partial^3 \partial^3 \partial^4 \partial^3 \partial^3 \partial^4 \partial^{\cdot} \partial^{\circ} \partial^3 \partial^4 \partial^{1/2} \partial^{1/2} \partial^{\circ} \tilde{N} \hat{I} \partial \cdot \partial^{1/2} \partial^{\circ} \partial \mu \hat{N} \hat{H}$
 $\partial \cdot \partial^{\circ} \tilde{N} \zeta \partial^{\circ} \partial \pm \partial^{\circ} \hat{N} \hat{H} \tilde{N} \hat{I} \partial^2 \partial^{\circ} \hat{N} \hat{H} \tilde{N} \hat{I} (\tilde{N} \acute{g} \partial \cdot \tilde{N} \zeta \partial^3 \partial^4 \tilde{N} \mathcal{H} \partial \cdot \tilde{N} \hat{I} \partial \cdot \tilde{N} \hat{H} \partial^3 \partial^4 \partial^{1/4}) ** .$

$\text{Tr} \hat{N} \text{Tr}^2 \text{Tr} \hat{N} :$
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